

CHAPTER 2: LITERATURE REVIEW

2.1. Introduction

A comprehensive literature review provides the foundation for developing new knowledge and guiding informed decision-making. In this study, the literature review examines previous works related to cybersecurity awareness, training frameworks, and strategies for enhancing employee behavior in SMEs. The review aims to identify both challenges and effective approaches that can ensure increased cybersecurity knowledge, reduced human error, and stronger organizational security, with a focus on application in SMEs in Cameroon.

Cybersecurity awareness is a critical component of organizational resilience. SMEs often face threats such as phishing, malware attacks, and social engineering, yet many lack structured awareness programs for employees (Kaspersky, 2021). Human error remains the leading cause of cybersecurity incidents, and without proper training, SMEs remain highly vulnerable (ENISA, 2020). Consequently, there is a growing interest in digital frameworks that combine education, interactive modules, and monitoring tools to mitigate cybersecurity risks effectively.

Cybersecurity awareness frameworks typically involve identifying potential threats, educating employees on safe practices, and providing continuous reinforcement through assessments and feedback. Research has shown that structured programs combining policy education, simulated attacks, and regular training updates are more effective than ad hoc training sessions (Alshaikh et al., 2020). By integrating technology with human-centered training, organizations can significantly reduce the likelihood of breaches caused by negligence or lack of knowledge.

Despite the proven benefits, SMEs in developing countries face challenges in implementing awareness frameworks. Financial constraints, limited access to cybersecurity expertise, and lack of organizational culture prioritizing security are common barriers (Okoro & Nwachukwu, 2020). Additionally, SMEs often operate with outdated IT infrastructure, which can complicate the deployment of advanced training tools and monitoring systems. Privacy and data protection are also critical concerns, as employee data must be handled securely during training exercises, simulations, and assessments.

The literature review draws on academic research, technical documentation, case studies, and industry best practices. Emphasis is placed on solutions that balance cost-effectiveness with technological reliability, such as web-based training platforms, modular frameworks, and scalable designs suitable for SMEs with limited resources.

In summary, this chapter highlights the importance of cybersecurity awareness in SMEs, reviews key strategies and technologies, and assesses how these can be adapted to the Cameroonian context. The insights gained will inform the design of a **Cybersecurity Awareness Framework**, ensuring it is effective, scalable, and tailored to local organizational realities.

2.2. Overview of SME Network and IT Infrastructure

Effective cybersecurity awareness depends on the IT environment in which employees operate. SMEs typically have smaller networks, fewer IT staff, and limited security infrastructure compared to large organizations. Their IT architecture often includes servers, workstations, internet connectivity, routers, firewalls, and sometimes cloud-based services. Ensuring that cybersecurity awareness programs align with this infrastructure is essential for practical and effective implementation (Ahmed et al., 2020).

The success of a cybersecurity awareness framework relies on stable IT systems to deploy training modules, conduct simulations, and monitor employee engagement. Unstable networks, outdated hardware, or unsupported software can hinder the delivery of interactive content and compromise the effectiveness of the program (Mbembe & Tchoumba, 2022).

Security considerations, such as firewalls, secure Wi-Fi, and access control, must also be incorporated to protect both the SME's operational data and employees' training records (Ola & Adegoke, 2021). The framework should be designed to operate under these constraints, using low-bandwidth-friendly web applications, offline access where possible, and simple integration with existing systems.

Globally, SMEs adopting well-structured awareness programs report improved cybersecurity posture, fewer breaches, and enhanced employee accountability (Fernandez et al., 2022). In Cameroon, integrating such frameworks into SMEs is both a strategic and operational

necessity to protect organizational assets, build employee competence, and reduce human-factor risks.

2.3. Challenges in Cybersecurity Awareness for SMEs

Cybersecurity awareness remains one of the most critical yet neglected aspects of SME management. Many SMEs continue to rely on informal or ad hoc approaches, leaving employees ill-equipped to handle cyber threats (ENISA, 2020). Common challenges include:

1. **Limited Knowledge and Skills:** Employees often lack understanding of phishing, social engineering, password hygiene, and safe browsing practices (Kaspersky, 2021).
2. **Time Constraints:** SMEs prioritize operational tasks, leaving little time for structured training programs.
3. **Financial Barriers:** Limited budgets restrict access to professional training platforms, simulations, and monitoring tools.
4. **Lack of IT Expertise:** Small organizations may lack IT personnel capable of implementing or managing awareness programs effectively.
5. **Unstable Infrastructure:** Poor internet connectivity, outdated devices, and unsupported software can hinder digital training delivery.

These challenges highlight the need for a cost-effective, adaptable, and scalable cybersecurity awareness framework tailored to the realities of SMEs in Cameroon. Effective programs can improve employee behavior, reduce the likelihood of breaches, and support organizational resilience.

2.4. Concepts of Redundancy and Resilience in Cybersecurity Awareness Frameworks

Redundancy and resilience are essential principles in designing sustainable cybersecurity awareness frameworks.

- **Redundancy** ensures multiple channels for training and monitoring. For example, web-based modules can be supplemented with downloadable content, printable guides, or offline assessments to maintain continuity in case of connectivity issues (Rahman et al., 2022).

- **Resilience** refers to the framework's ability to adapt, recover, and continue delivering training despite challenges such as network outages, hardware failures, or staff turnover. Resilient systems allow employees to resume learning without losing progress or critical data.

Common strategies include multiple learning platforms (online, offline, and hybrid), cloud-based storage of training progress, alternative assessment methods, and automated alerts to administrators in case of missed training or system failures. Security remains central; employee data, assessment scores, and simulated phishing responses must be encrypted and stored securely (Sharma et al., 2021).

Implementing redundancy and resilience ensures continuous employee engagement, reliable progress tracking, and the long-term success of cybersecurity awareness initiatives, even in low-resource SME environments.

2.5. Technologies Used in Cybersecurity Awareness Frameworks

Effective frameworks integrate hardware, software, and monitoring tools to deliver training and assess employee behavior. Key technologies include:

1. **Web-Based Learning Platforms:** Accessible from desktops, laptops, and mobile devices to deliver modules and quizzes.
2. **Learning Management Systems (LMS):** Centralized platforms to track employee progress, generate reports, and provide automated reminders.
3. **Simulation Tools:** Phishing simulations, social engineering exercises, and interactive scenarios to reinforce practical learning.
4. **Database Management Systems:** Secure storage for user accounts, training results, and activity logs. Open-source systems like MySQL or PostgreSQL can reduce costs.
5. **Cloud Infrastructure:** Enables scalable access, secure backups, and analytics capabilities for monitoring employee engagement.
6. **User Interface (UI) Tools:** Dashboards and mobile apps that present progress, scores, and alerts in a clear, user-friendly manner.
7. **Security Measures:** Encryption, role-based access control, and regular updates protect sensitive data related to employees and SMEs.

When deployed effectively, these technologies improve training accessibility, increase retention of cybersecurity knowledge, and support organizational resilience.

2.6. Case Studies on Cybersecurity Awareness Frameworks

Case Study 1: Small Business Awareness Program in Nigeria

A group of SMEs implemented a structured cybersecurity awareness framework combining online modules and phishing simulations. Employees were trained on email safety, password management, and device security. Post-training assessments showed a 70% reduction in risky behaviors, demonstrating the effectiveness of structured awareness programs even with limited resources.

Case Study 2: Cloud-Based Awareness Framework in Kenya

Several SMEs adopted cloud-hosted LMS platforms to deliver interactive cybersecurity training. Modules included real-time quizzes, reporting dashboards, and automated notifications. SMEs reported improved compliance with security policies and faster incident reporting, even in low-bandwidth environments.

Case Study 3: Hybrid Training for SMEs in South Africa

A hybrid model combining in-person workshops and online simulations was deployed across multiple small enterprises. Redundancy in training delivery (online + offline) ensured continuity despite internet outages. Employees displayed improved engagement and responsiveness to simulated phishing attacks.

Lessons Learned:

- Phased deployment and pilot programs reduce adoption barriers.
- Redundancy (multiple access channels) ensures continuity.
- Interactive and practical exercises increase retention.
- Security of training data is critical to maintain trust.

2.7. Security Considerations in Framework Design

The framework must embed security at every level:

1. **Encrypted Data Transmission:** Use SSL/TLS for all platform communications.
2. **Role-Based Access Control:** Restrict access to administrators, trainers, and relevant staff.
3. **Regular Updates:** Keep LMS software, modules, and servers patched against vulnerabilities.
4. **Monitoring and Alerts:** Track unauthorized access attempts or unusual activity.
5. **Secure Backup:** Regular cloud and local backups to prevent data loss.
6. **Audit Trails:** Log activities to monitor progress and detect anomalies.
7. **Disaster Recovery Drills:** Simulate outages or cyber incidents to ensure framework resilience.

2.8. Summary of Key Insights

1. Employee awareness is critical for reducing human-factor cybersecurity risks.
2. Redundancy and resilience in training delivery ensure continuity.
3. Web-based platforms, cloud storage, and simulation tools enhance learning and monitoring.
4. Security of training data and assessment results is essential.
5. Limited resources, unstable infrastructure, and lack of expertise present challenges.
6. Best practices include phased implementation, hybrid learning modes, and practical exercises.

2.9. Research Gaps

1. Limited studies on SMEs in Cameroon regarding cybersecurity awareness.
2. Few frameworks adapted to low-resource environments with unstable IT infrastructure.
3. Scarce practical deployment models for small organizations.
4. Lack of integrated models combining training, monitoring, and security of employee data.
5. Insufficient focus on maintenance and training for non-technical staff.

2.10. How This Study Addresses the Gaps

1. Local adaptation for SMEs in Cameroon considering resource constraints.

2. Cost-effective technologies using open-source platforms and lightweight databases.
3. Step-by-step implementation roadmap with pilot testing and phased rollout.
4. Integrated security and privacy measures for employee data.
5. Scalable and maintainable framework suitable for SMEs with minimal ICT expertise.
6. Alignment with business objectives, improving compliance, resilience, and overall cybersecurity culture.